Veluz, Cliff Richard R.
BSIT - 4B
ITEP 414

Week 2 Portfolio Project

Part 1: Company Profile

Company Name: ABC Startup Solutions
Nature of Business: Software Development
Company Vision: To become a leading provider of innovative, scalable, and secure software applications that drive digital transformation for businesses worldwide.
Company Mission: To engineer high-quality software products from the ground up, fostering a collaborative environment where technical excellence and creative problem-solving thrive.
Office Location (fictional): 101 Bay, Laguna Suite 300 (Single Floor)
Organizational Structure: Flat Management Structure

- Chief Executive Officer (CEO)
- Department Heads (IT, HR, Finance, Sales)
- Individual Contributors / Specialists

Employee Distribution (20 Total):

- Information Technology: 5
- Human Resources: 4
- Finance: 5
- Sales: 6

Part 2: Enterprise Hardware Inventory

Asset ID	Hardware	Quantity	Department	Purpose
HW-DESK-01	Desktop Computers (Standard)	9	HR (4), Finance (5)	To provide secure, stationary workstations for processing sensitive employee and financial data.
HW-LAPT-01	Laptops (High-Performance)	5	IT (5)	To handle resource-heavy tasks like mobile systems development, compiling code,

				and running local database management queries.
HW-LAPT-02	Laptops (Standard)	6	Sales (6)	To enable mobility for client meetings, presentations, and remote communication.
HW-SERV-01	Enterprise Server	1	IT / Entire Company	To act as the central domain controller, manage local DNS/DHCP, and host internal testing environments.
HW-ROUT-01	Edge Router / Firewall	1	IT / Entire Company	To provide secure internet access, manage core routing, and protect the network from external threats.
HW-SWIT-01	24-Port PoE+ Switch	1	IT / Entire Company	To physically connect all desktop computers, the server, and power the wireless access points.
HW-PRIN-01	Enterprise Multi-function Printer	1	HR, Finance, Admin	To handle essential document printing, scanning, and copying.
HW-UPS-01	Uninterruptible Power Supply (UPS)	2	IT / Server Room	To provide emergency backup power and surge protection to the server, router, switch, and NAS.
HW-WAP-01	Wireless Access Point (Wi-Fi 6)	2	IT / Entire Company	To ensure seamless wireless

				internet coverage across the office for all 11 laptops and mobile devices.
HW-NAS-01	NAS Storage (4-Bay)	1	IT / Entire Company	To serve as a centralized repository for large assets, e-commerce platform staging files, and automated network backups.
HW-BACK-01	External Backup Drive (8TB)	2	IT	To facilitate offline, air-gapped backups of the NAS that can be rotated off-site weekly for disaster recovery.
HW-MONI-01	27-inch LED Monitors	20	All Departments	To serve as primary displays for the 9 desktops and secondary productivity displays for the 11 laptops.

Explanation:

Workstations (Desktops/Laptops): I distributed the workstations based on departmental needs. HR and Finance deal with sensitive data and don't require mobility, making desktops the cost-effective and secure choice. The IT team needs high-performance laptops with extra RAM to efficiently run emulators for mobile applications and manage heavy database loads. Sales requires standard laptops for travel and client pitches.
Networking (Router, Switch, WAPs): A single enterprise router paired with a 24-port switch is sufficient to handle our 9 hardwired desktops, server, and NAS. Two Wireless Access Points will easily cover a single office floor to support the 11 laptops.
Storage & Redundancy (Server, NAS, UPS, Backup Drives): While the server handles active directory and access management, the NAS is dedicated to bulk file storage (like UI/UX assets or e-commerce backups). The two external drives allow for a safe, rotating off-site backup strategy. The UPS units are critical to ensure our infrastructure doesn't immediately crash or corrupt during a power flicker.
Peripherals (Monitors & Printer): Every employee gets one external monitor for ergonomic productivity. A single, high-capacity central printer is highly efficient for a 20-person office, reducing maintenance overhead.

Part 3: Enterprise Software Inventory

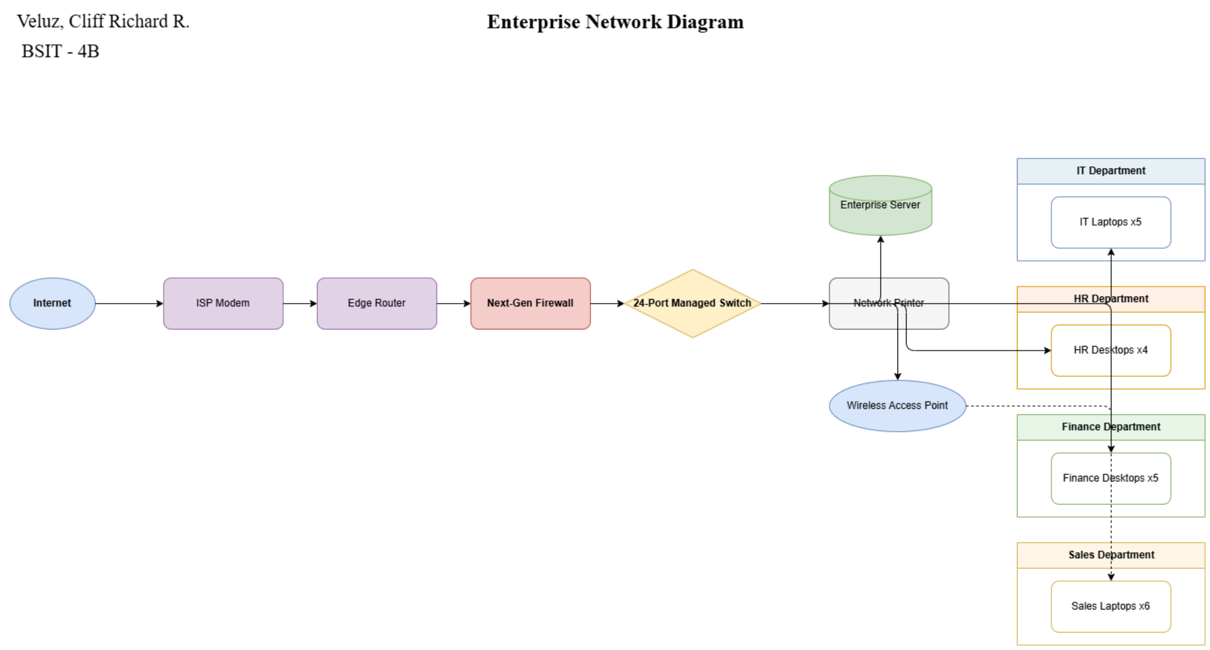
Software	Version	License	Purpose
Windows 11 Pro	23H2	OEM / Volume Licensing	Primary operating system for all 20 employee laptops and desktop workstations.
Ubuntu Server	24.04 LTS	Open Source (GPL)	Primary operating system for the on-premise Enterprise Server.
Microsoft Office	Microsoft 365 Apps	Commercial Subscription	Standard productivity suite for all departments.
VS Code	Latest Stable	Freeware (MIT)	Primary Integrated Development Environment (IDE) for the IT department.
Git	2.45 (or latest)	Open Source (GPL)	Command-line version control system for source code management.
GitHub Desktop	Latest Stable	Freeware (MIT)	Graphical User Interface (GUI) for streamlining repository management.
VirtualBox	7.0 (or latest)	Open Source (GPLv2)	Desktop virtualization software for running isolated virtual machines.
Google Chrome	Latest Enterprise	Freeware	Standardized web browser for all employees.
Microsoft Defender	Integrated	Included with OS	Baseline endpoint security and antivirus protection.
AnyDesk	Latest Stable	Commercial Subscription	Remote access and support utility.
7-Zip	24.05 (or latest)	Open Source (LGPL)	File archiving and compression utility.

Part 4: Enterprise Network Inventory

Equipment	Quantity	Specification / Description	Purpose
ISP Modem	1	Gigabit Fiber Modem (ISP Provided)	Serves as the demarcation point, connecting ABC Startup Solutions' internal network to the wider internet.
Router	1	Enterprise Gigabit VPN Router	Routes traffic between our Local Area Network (LAN) and the internet (WAN), managing IP addresses and network traffic flow.
Firewall	1	Next-Generation Hardware Firewall (NGFW)	Inspects all incoming and outgoing traffic, actively blocking unauthorized access and enforcing company security policies.
Switch	1	24-Port Gigabit PoE+ Managed Switch	Physically connects all wired devices (desktops, server, NAS, printer) and provides Power over Ethernet (PoE) to the access points.
Access Point	2	Wi-Fi 6 (802.11ax) Dual-Band Access Points	Broadcasts a fast, secure wireless network across the single office floor to support the 11 laptops and authorized mobile devices.
Patch Panel	1	24-Port CAT6 Rackmount Patch Panel	Organizes and terminates the structural cable runs from the office walls into the central server rack, keeping connections tidy.
CAT6 Cables	1 Box	1000 ft Bulk Spool (Solid Copper)	Provides high-speed, reliable physical

			wiring for wall drops and ceiling runs connecting office endpoints back to the server room.
RJ45 Connectors	1 Pack	100-Pack CAT6 Connectors	Used to terminate the cut ends of the bulk CAT6 cable so they can be plugged into wall jacks, devices, and the patch panel.

Part 5: Enterprise Network Diagram



Part 6: System Administration Roles

Helpdesk Technician

- **Responsibilities:** Acting as the first line of defense for IT support, Helpdesk Technicians are responsible for answering support requests, troubleshooting basic hardware and software issues, resetting passwords, and setting up workstations for new employees. They manage the flow of IT tickets and escalate complex issues to specialized administrators.
- **Skills:** Exceptional communication and customer service skills, patience, strong problem-solving abilities, and a foundational understanding of operating systems (Windows, macOS) and basic networking.

- **Common Tools:** Ticketing systems (Jira Service Management, Zendesk, ServiceNow), Remote desktop software (AnyDesk, TeamViewer), and Active Directory for basic user management.
- **Certifications:** CompTIA A+, ITIL Foundation, Microsoft 365 Certified: Endpoint Administrator Associate.

Network Administrator

- **Responsibilities:** Network Administrators design, configure, and maintain the organization's Local Area Network (LAN) and Wide Area Network (WAN). They manage routers, switches, and firewalls, monitor network performance and traffic for bottlenecks, and ensure stable and secure connectivity for all users.
- **Skills:** Deep knowledge of network protocols (TCP/IP, DNS, DHCP), subnetting, routing protocols (OSPF, BGP), and a strong grasp of network security principles (VPNs, VLAN segregation, firewall rules).
- **Common Tools:** Network monitoring software (SolarWinds, PRTG), Packet analyzers (Wireshark), Terminal emulators (PuTTY), and hardware management interfaces (Cisco IOS).
- **Certifications:** Cisco Certified Network Associate (CCNA), Cisco Certified Network Professional (CCNP), CompTIA Network+.

Linux System Administrator

- **Responsibilities:** These administrators are responsible for the installation, configuration, and maintenance of Linux-based servers. Their duties include managing user permissions, performing routine system updates, securing server environments, troubleshooting system errors, and writing scripts to automate repetitive tasks.
- **Skills:** High proficiency in the Linux Command Line Interface (CLI), strong shell scripting abilities (Bash, Python), understanding of Linux file systems, and experience with server hardening and patch management.
- **Common Tools:** SSH for remote access, automation and configuration management tools (Ansible, Puppet), containerization platforms (Docker), and text editors (Vim, Nano).
- **Certifications:** Red Hat Certified System Administrator (RHCSA), CompTIA Linux+, Linux Foundation Certified System Administrator (LFCS).

Cloud Administrator

- **Responsibilities:** Cloud Administrators manage the organization's cloud-based infrastructure (e.g., AWS, Azure, Google Cloud). They are responsible for provisioning virtual servers, managing cloud storage, monitoring resource utilization to control billing costs, and implementing disaster recovery and backup solutions within the cloud environment.
- **Skills:** Understanding of virtualization, cloud networking, Identity and Access Management (IAM), cloud security best practices, and familiarity with Infrastructure as Code (IaC) principles.
- **Common Tools:** Cloud provider dashboards (AWS Management Console, Azure Portal), Infrastructure as Code tools (Terraform, AWS CloudFormation), and container orchestration (Kubernetes).
- **Certifications:** AWS Certified SysOps Administrator, Microsoft Certified: Azure Administrator Associate, Google Cloud Associate Cloud Engineer.

➢ **How these four professionals work together inside one organization.**

- In a company these four experts operate within a very well-coordinated system for handling issues and providing support. The Helpdesk Technician is the person users see first in the IT department. This person deals with problems that users face and looks for bigger issues that need to be passed on. If a helpdesk ticket shows that users can't access an internal program the Network Administrator gets involved. This person looks into how data moves through the network checking the cables, switches and security systems to make sure everything is working right. At the time the Linux System Administrator looks at the servers that run the program making sure the operating system is working, the services are running and there is enough space and power. If the company has a mix of, on-site and setups or if everything is remote the Cloud Administrator joins the team. This person makes sure the virtual networks, the tools that spread the workload and the cloud computers are all set up right, safe and working with the rest of the company's systems.

References:

- Bureau of Labor Statistics, U.S. Department of Labor. (2025). *Computer Support Specialists*. Occupational Outlook Handbook. **https://www.bls.gov/ooh/computer-and-information-technology/computer-support-specialists.htm**
- Bureau of Labor Statistics, U.S. Department of Labor. (2025). *Network and Computer Systems Administrators*. Occupational Outlook Handbook. **https://www.bls.gov/ooh/computer-and-information-technology/network-and-computer-systems-administrators.htm**
- CompTIA. (2024). *What Is a Linux Administrator?* CompTIA IT Career News. **https://www.comptia.org/en-us/blog/your-next-move-linux-administrator/**
- Koenig Solutions. (2023). *Red Hat System Administrator Roles and Responsibilities, Job Description & Salary*. **https://www.koenig-solutions.com/blog/red-hat-system-administrator-roles-and-responsibilities-job-description-salary**
- Koenig Solutions. (2023). *Microsoft Azure Administrator Roles and Responsibilities*. **https://www.koenig-solutions.com/blog/microsoft-azure-administrator-roles-and-responsibilities**

Part 7: Infrastructure Recommendations

To ensure ABC Startup Solutions operates securely, efficiently, and with high availability, the following infrastructure recommendations have been tailored for our 20-employee software development environment:

1. **Internet Service Provider (ISP)**

- **Recommendation:** A Dedicated Internet Access (DIA) line from a leading local enterprise provider, such as **PLDT Enterprise (iGate)** or **Converge ICT (Enterprise DIA)**.

Specifications: Minimum 500 Mbps symmetrical download/upload, a static public IP address, and a guaranteed Service Level Agreement (SLA) of 99.9% uptime.

2. Server Specifications & Environmental Considerations

- Recommendation: 1U Rackmount Server (e.g., Dell PowerEdge R450 or HPE ProLiant DL320 Gen11, which are easily sourced through local Philippine distributors like VST ECS or WSI).
  - **Processor:** Intel Xeon Silver or AMD EPYC (8 cores / 16 threads)
  - **RAM:** 64GB DDR4/DDR5 ECC Memory
  - **Storage:** 4 x 2TB Enterprise NVMe/SATA SSDs configured in RAID 5 (usable storage approximately 6TB with 1-drive fault tolerance)
  - **Power & Cooling :** Redundant Dual Power Supplies (PSUs) connected to heavy-duty Double-Conversion UPS (Uninterruptible Power Supply) units (e.g., APC or Eaton).
- **Logical Justification:** The hardware itself provides the necessary compute power and fault tolerance (RAID 5) for local database hosting and internal testing. However, the Philippine grid can be prone to sudden power fluctuations, brownouts, and rolling blackouts (especially during the summer months or typhoon season). Double-conversion UPS units are mandatory to provide clean, continuous power and prevent hardware degradation or data corruption during grid transitions.

3. Backup Strategy

- **Recommendation:** Implementation of the **3-2-1 Backup Rule**:
  - **3 Copies of Data:** One primary production copy, one local backup, and one off-site backup.
  - **2 Different Media Types:** Local Network-Attached Storage (NAS) for fast hourly snapshots, and encrypted external hard drives for weekly air-gapped storage.
  - **1 Copy Stored Off-site:** Encrypted, immutable cloud backup repository (e.g., AWS S3 Glacier or Microsoft Azure, utilizing their Southeast Asia server regions like Singapore for lower latency).
- **Logical Justification:** This strategy guarantees business continuity in the event of hardware failure, ransomware infection, or physical disasters (such as flooding or severe typhoons which can occasionally impact parts of Laguna). Air-gapped physical drives offer total protection against network-based ransomware.

4. Security Recommendations

- **Recommendation:**
  - **Network Segmentation (VLANs):** Split the office network into isolated Virtual Local Area Networks (Management, IT, HR/Finance, Sales, and Guest Wi-Fi).
  - **Full Disk Encryption:** Enforce BitLocker on all Windows endpoints.
  - **Multi-Factor Authentication (MFA):** Enforce mandatory MFA across all corporate email accounts, VPN access, and administrative dashboards.
- **Logical Justification:** Segmentation prevents lateral movement in the event of a security breach (e.g., a guest in the lobby cannot access the Finance VLAN). BitLocker is especially important in the Philippines to protect proprietary source code and confidential HR data if a laptop is lost or stolen while an employee is commuting.

5. Antivirus & Endpoint Protection

- **Recommendation:** Centralized Next-Generation Antivirus (NGAV) / Endpoint Detection and Response (EDR) utilizing **Microsoft Defender for Business** (integrated via Microsoft 365).
- **Logical Justification:** Centralized cloud management allows the system administrator to monitor security alerts across all 20 devices from a single console. It provides enterprise-grade protection without requiring the purchase of expensive third-party endpoint licenses, which is highly cost-effective for a local startup.

6. Password Policy

- **Recommendation:** Enforce modern security standards through Active Directory / Group Policy:
    - **Length:** Minimum 14 characters.
    - **Account Lockout:** Automatic lockout after 5 consecutive failed attempts.
    - **Credential Manager:** Provide a standardized corporate password manager (e.g., Bitwarden / 1Password).
- **Logical Justification:** Aligns with modern global NIST (National Institute of Standards and Technology) guidelines. Providing a password manager eliminates the common local practice of writing passwords on sticky notes attached to monitors, drastically improving operational security.

7. Expansion Plan

- **Recommendation:** Scalability-ready hardware and architectural provisioning:
    - **Switching Capacity:** The chosen 24-port switch leaves open ports for near-term growth, with the server rack designed to accommodate an additional 24- or 48-port switch later.
    - **Cloud Hybrid Transition:** Structure local Active Directory to synchronize seamlessly with Microsoft Entra ID.
- **Logical Justification:** Laguna's tech ecosystem is growing rapidly. This hybrid setup allows the company to seamlessly transition to a "Work From Anywhere" model, easily onboarding remote developers from Metro Manila or other provinces without having to constantly expand the physical Laguna office space or purchase new physical servers.

Part 8: Personal Reflection

I learned that being a system administrator professional requires more than just repairing malfunctioning computer systems and networks. It requires having the knowledge of building reliable and scalable infrastructures that empower businesses through meeting organizational objectives by enabling all the critical departments to deliver their services efficiently and securely. Through this activity, I learned how to set up computer hardware configurations, set up appropriate software platforms, and design networks that facilitate reliable communication and data exchange in an organization.

The designing of the enterprise network topology and hardware acquisition was challenging for me because it did not seem as easy as I expected it to be. Coming up with specifications for the hardware components such as the number of computers and switches needed to set up the network was a challenging task. Making sure there would be enough cabling and ports to support the network topology and that network traffic would move safely between various departments, like the finance and sales departments, presented me with a number of difficulties. In order to guarantee dependability and security in the communications infrastructure, I had to pay attention to these nuances of network design and management, even though there were numerous ways to deploy the hardware equipment and network topology.

Therefore, the project helped me understand the importance of making appropriate infrastructure plans before putting the technology solutions into an organization. Planning is often the most important step in the implementation of new information technology systems. If acquisition plans are not properly developed and followed, the organization may incur unnecessary expenses in the acquisition of hardware and software. And even worse, when a company invests in the purchase of obsolete technology products, there are no guaranteed results. A good infrastructure planning process will also reduce risks related to power outages and data security and will also provide suggestions on how to implement systems. From this activity, I have gained knowledge on how a system administrator works and its role. This activity will help me be more confident and innovative to deal with real world IT problems as a student. It has pushed me to go beyond the theoretical side of IT infrastructure planning and its practical implications.